

Chapter 40. Care of the Mechanically Ventilated Patient with a Tracheotomy

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Care of the Mechanically Ventilated Patient with a Tracheotomy: Introduction

Although Egyptian tablets depict use of tracheotomy for medical applications nearly 5600 years ago,¹ initial descriptions of the procedure in Western literature did not appear until the middle of the sixteenth century.² By 1718, “tracheotomy” became accepted terminology for the surgical technique that was then primarily used for relief of airway obstruction and removal of aspirated foreign bodies. Typically gruesome clinical results relegated tracheotomy to a reviled role in airway management and gained it a designation as the “scandal of surgery.”³ The diphtheria epidemics of the nineteenth century popularized tracheotomy.⁴ Tracheotomy did not become widely accepted, however, until 1909, when Chevalier Jackson standardized surgical techniques and decreased the operative mortality from 25% to less than 1%.⁵

Advances in tube design during the 1960s and 1970s and improved management techniques further promoted acceptance of tracheotomy for long-term airway access for critically ill, ventilator-dependent patients. The advent of percutaneous dilational tracheotomy (PDT) further widened the application of tracheotomy by allowing the procedure to be performed in the intensive care unit (ICU) by nonsurgeons.⁶ Up to 24% of patients undergoing mechanical ventilation and 6% of critically ill patients in general have a tracheotomy performed.^{6–12} In North Carolina, nearly a threefold increase in the application of tracheotomy for prolonged mechanical ventilation was observed from 1993 to 2002.⁹ Although only 7% of ventilated patients underwent tracheotomy in that study, they accounted for 22% of all mechanical ventilation patient charges.

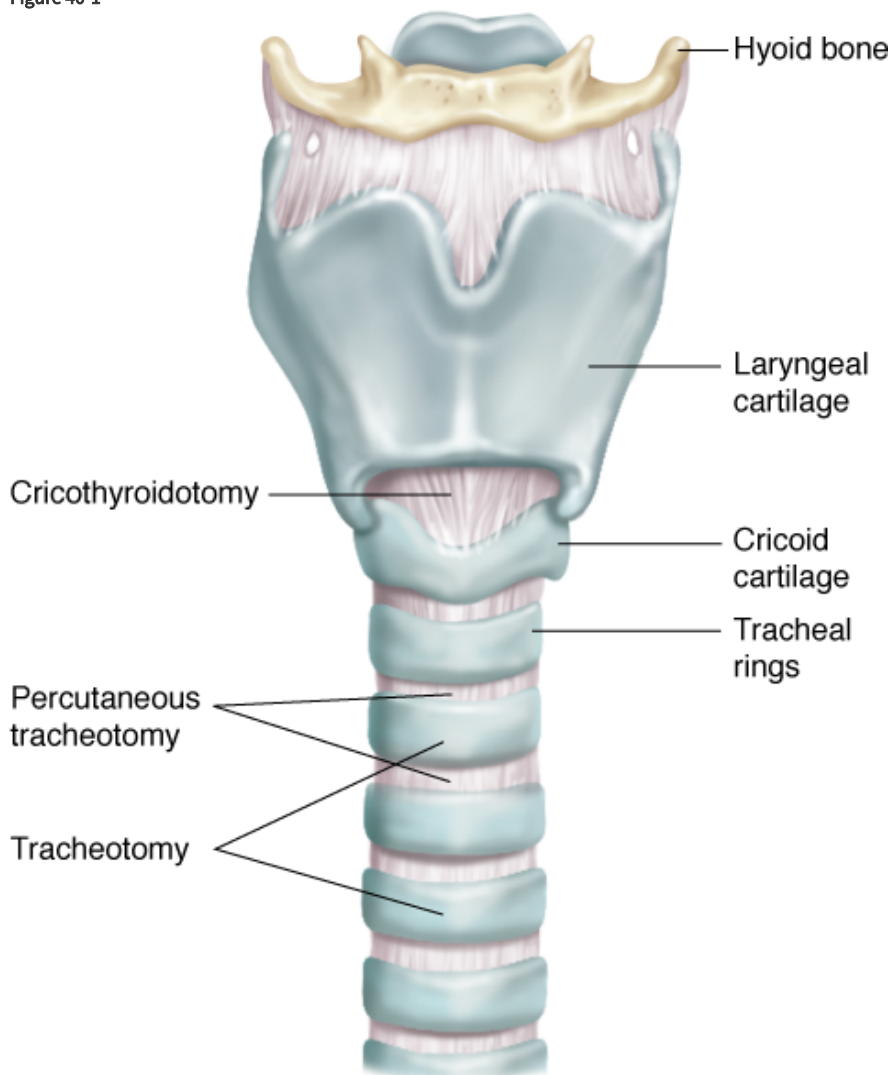
Four indications exist for a tracheotomy in critically ill patients: (a) maintenance of airway patency for patients with functional or mechanical upper airway obstruction, (b) provision of airway access for suctioning retained airway secretions, (c) prevention or limitation of aspiration in patients with glottic dysfunction, and (d) management of patients who require long-term airway access for ventilator support.¹³ Acceptable outcomes from tracheotomy remain dependent on the skill of the operator who performs the procedure and expertise of interdisciplinary teams charged with managing patients from the critical care phase of their illnesses through transitions to hospital wards and long-term care facilities until successful decannulation occurs.^{14,15}

Techniques of Surgical Airway Access

Standard Surgical Tracheotomy

A *standard surgical tracheotomy* provides tracheal access through a temporary incisional tracheostoma between cartilaginous rings (Fig. 40-1). The stoma spontaneously closes after removal of the tracheotomy tube.

Figure 40-1



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Anatomic location of stoma placement for different forms of surgical airway access.

Standard surgical tracheotomy is a well-tolerated procedure with an operative mortality of less than 1% when performed as an elective procedure in stable, ventilator-dependent patients. In contrast, most,¹⁶ but not all,¹⁷ centers report a higher complication rate when surgical tracheotomy is performed as an emergency procedure. Some groups report successful emergency tracheotomy when performed on awake patients.¹⁸ Most centers, however, have replaced emergency surgical tracheotomy with specialized endotracheal intubation techniques to secure a difficult airway,¹⁹ emergency cricothyroidotomy,²⁰ and emergency percutaneous techniques.^{21–27}

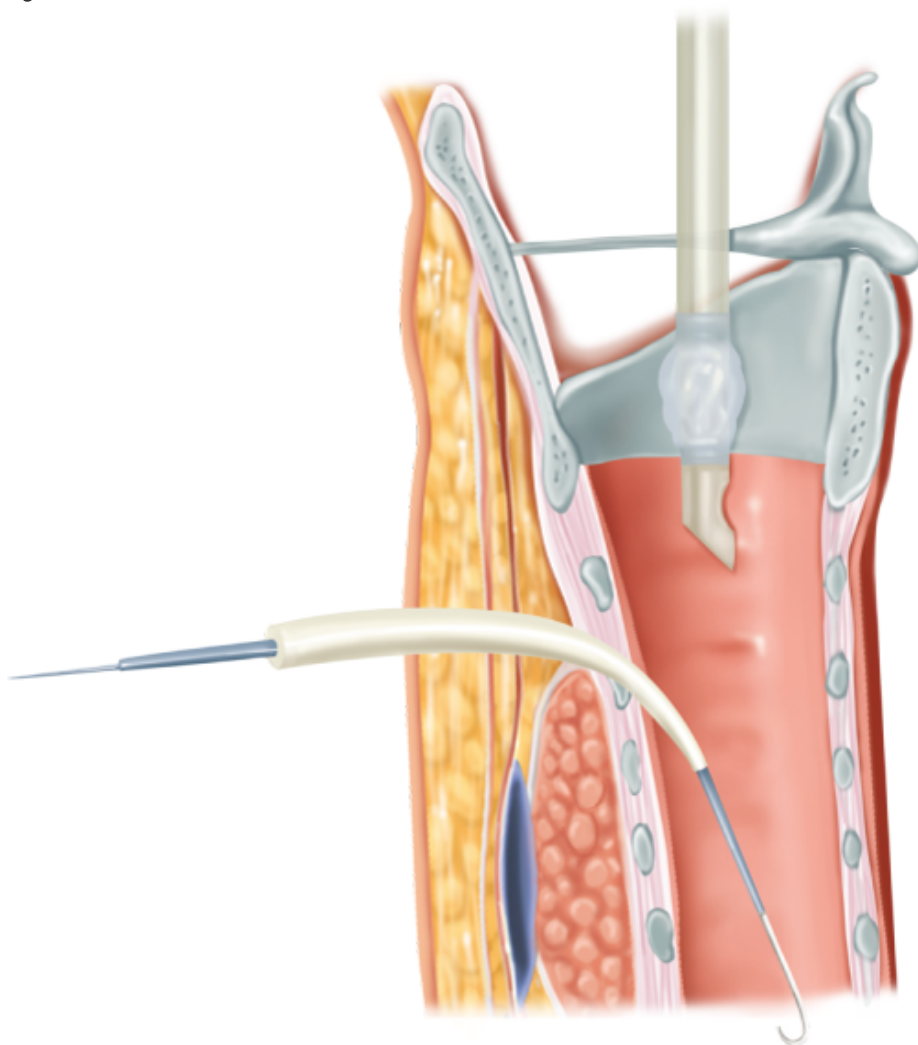
Percutaneous Tracheotomy

Percutaneous tracheotomy refers to several techniques that insert a standard or modified tracheal airway with a Seldinger technique below the first or second tracheal rings using a device to cut and spread the trachea²⁸ or a forceps

or dilator technique to cannulate and dilate tracheal tissue between cartilaginous rings (see Fig. 40-1).²⁹⁻³¹

Ciaglia first described percutaneous dilatational tracheotomy (PDT) wherein a Seldinger technique allowed the insertion of sequential dilators to place a tracheotomy tube.³⁰ The procedure has evolved from the use of sequential dilators to insertion of a single dilator that increases in caliber from tip to base where it matches the diameter of a tracheotomy tube (Ciaglia Blue Rhino, Cook Critical Care Inc., Bloomington, IN) (Fig. 40-2).³² The single-dilator Blue Rhino technique results in faster insertion times as compared with the sequential dilator technique,^{27,33,34} and a lower risk of posterior tracheal injury.^{27,35,36}

Figure 40-2



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Insertion of a stoma dilator for percutaneous dilatational tracheotomy. (Courtesy Cook Critical Care, Bloomington, IN.)

Recently, the Ciaglia PDT has been further modified with use of a balloon dilator (Ciaglia Blue Dolphin, Cook Critical Care Inc., Bloomington, IN), which is placed across the anterior tracheal wall and inflated to dilate the stoma by radial force, thereby avoiding anterior-posterior compression of tracheal structures that can fracture cartilaginous rings. The procedure has not yet been established as being superior to the Blue Rhino technique.^{37,38}

The Ciaglia PDT technique is the most commonly performed percutaneous tracheotomy procedure in the United States and is frequently used in Europe.^{39–41} Nonsurgeons can perform the procedure with low complication rates, with few patients requiring conversion to a surgical tracheotomy.⁴² Controversy remains, however, among otolaryngologists, with only 29% of otolaryngology residency programs teaching PDT; however, programs that do include training in PDT use the Ciaglia Blue Rhino technique and endorse its safety.^{43,44}

Other techniques for percutaneous tracheotomy include a specialized guidewire dilating forceps with a groove that allows loading of a guidewire onto forceps, which dilate the trachea, thread the wire, and insert a tracheotomy tube by Seldinger technique (Portex guidewire dilating forceps kit, Sims, Inc, Philadelphia, PA).⁴⁵ The guidewire dilating forceps technique provides quick, safe, and effective placement of a tracheotomy for critically ill patients.^{46–51} Guidewire dilating forceps require the same time to completion as the Ciaglia single-dilator method.^{52,53} Comparative studies have shown both higher,⁵⁴ similar,^{46,52,53} and lower⁵⁰ complication rates with the guidewire dilating forceps as compared with the Ciaglia Blue Rhino technique, with each having unique complications.⁵⁵ Long-term follow-up studies demonstrate acceptable rates of tracheal stenosis and other chronic airway complications with guidewire dilating forceps,^{56–59} although one study of 208 patients observed a 38% incidence of a changed voice and a 12% incidence of persistent, severe cough. Patients with short, fat necks, conditions that prevent neck extension, an enlarged thyroid isthmus, a previous tracheotomy, coagulopathy, or anticoagulation therapy have successfully undergone the guidewire dilating forceps technique.⁶⁰

Frova et al described a single-step technique (PercuTwist, Rusch, Kern, Germany) that threads a screw-type dilator between tracheal rings to allow insertion of a specially designed tube.⁶¹ This technique causes less compression of the anterior tracheal wall than does the Ciaglia technique. Early experience with PercuTwist reported variable success with difficulties in learning the procedure and problems with damage to tracheal structures.^{34,62–65} More recent reports using bronchoscopic guidance compared PercuTwist with other percutaneous techniques and reported similar complication rates and speed of completion of the procedures.^{66–68}

Fantoni proposed translaryngeal tracheotomy for placing a tracheotomy in a reverse direction from within the airway through the tracheotomy tract.⁶⁹ The operator places a needle into the trachea between the second and third tracheal rings. With bronchoscope guidance, a guidewire is passed retrograde through the needle into a cuffed, rigid bronchoscope, which is then removed from the airway. A tracheotomy tube with a tapered proximal end is loaded onto the cephalad end of the guidewire and pulled through the airway and the needle tract. Comparative studies suggest that translaryngeal tracheotomy is equally effective and less traumatic than the Ciaglia PDT technique.^{70–73}

All percutaneous techniques require a learning curve to gain skills for completing the procedures quickly and safely.^{74,75} Centers that emphasize close supervision of trainees note no differences in complication rates by seniority of operator.⁷⁶ Because PDT is the primary technique used in most ICUs, it is the percutaneous technique discussed in the remainder of this chapter.

Contraindications to PDT have decreased with increasing experience with the procedure. Conservative relative contraindications have included age younger than 16 years, bleeding diatheses, severely calcified tracheal rings, and anatomic abnormalities, such as obesity and thyromegaly, that obscure landmarks.^{35,77–79} Greater experience with

PDT, however, has allowed its application by experienced practitioners in patients with body mass indices equal to or greater than 27 kg/m²⁸⁰ and equal to or greater than 35 kg/m².⁸¹ Preoperative ultrasonography allows selection of patients with anatomic abnormalities of the neck for successful PDT.^{82,83} Trauma and burn patients with known or suspected cervical injury, patients 5 to 6 days after anterior spinal fusion, and patients with previous tracheostomies have also undergone successful PDT.^{83–89} Studies report successful PDT for patients with thrombocytopenia,^{90,91} coagulopathies secondary to liver disease,⁹² and systemic heparinization.^{91,93} PDT has been performed successfully for patients with acute respiratory failure and dependence on high levels of positive end-expiratory pressure (mean: 16.6 cm H₂O; range: 12 to 20 cm H₂O positive end-expiratory pressure).⁹⁴

Purported advantages of PDT as compared with standard surgical tracheotomy include speed of the procedure (less than 10 minutes), ICU placement without operating room transport, avoidance of general anesthesia, decreased cost, reduced operating room use, and limited personnel requirements that include a single operator, a bronchoscopist, a nurse, and a respiratory therapist.^{95,96} Such simplifications decrease costs,⁹⁷ avoid complications from intrahospital transport, and eliminate dependency on the operating room schedule, thereby decreasing the time from the decision to place a tracheotomy to when it is actually performed.⁹⁸ Some centers, however, obviate some of these benefits by performing standard surgical tracheotomy in the ICU.^{99–109} Standard surgical tracheotomy in the ICU has similar costs and complication rates comparable to PDT if performed under controlled circumstances with all of the resources that would be available in the operating room.^{108,109}

Percutaneous tracheotomy has a potential advantage over standard surgical tracheotomy by dilating rather than incising cervical tissues thereby producing a tracheostoma that fits snugly around the tracheotomy tube.¹¹⁰ This tight fit has potential for improving anchoring of the tracheotomy tube, decreasing the incidence of inadvertent extubation (1.4% incidence),⁴³ compressing blood vessels in the stoma tract thus decreasing postoperative bleeding (2% to 4% incidence),^{35,43,111} and diminishing the incidence of wound infections observed with conventionally placed surgical stomas.¹¹¹ The smaller skin incision also may result in a better cosmetic appearance after extubation.^{111,112}

Many observational studies and a smaller number of randomized, controlled trials^{73,86,95,96,98,104,111–119} have addressed the potential benefits of PDT by comparing the outcomes of the procedure with those of standard tracheotomy. These trials, however, are relatively small, often employ varying tracheotomy techniques, have considerable spectrum bias, usually exclude patients with relative complications to PDT, and measure heterogeneous outcomes with differing definitions between studies. Five meta-analyses have critically analyzed these studies and come to varying conclusions.^{109,120–123} The meta-analysis by Dulguerov et al concluded that PDT had a higher complication rate than standard tracheotomy, but the analysis suffered from methodological flaws (double counting) and included retrospective studies.¹⁰⁹ The three most recent meta-analyses^{109,122,123} have the usual limitations from heterogeneity of primary studies, but conclude that PDT and surgical tracheotomy have comparable major complication rates. The analyses differ regarding relative rates of early postoperative complications, but Delaney et al performed a subgroup analysis by type of PDT and found a similar rate of postoperative bleeding for PDT as compared with surgical tracheotomy performed in the operating room.¹²³ General conclusions from these analyses indicate that the choice of technique should be individualized by an interdisciplinary approach and PDT can be performed more quickly with similar outcomes as compared with surgical tracheotomy performed in the operating room or at the

bedside in appropriately selected patients.¹⁰⁹ A shorter time from the decision to perform tracheotomy to its actual completion may result from PDT.^{98,111} Prospective comparative data of surgical versus PDT for patients with underlying bleeding diatheses, neck anatomic abnormalities, and other risk factors for complications do not exist.¹²⁴ Also, up to 7% of PDT procedures require conversion to surgical tracheotomy because of technical difficulties.¹²²

Cricothyroidotomy

Cricothyroidotomy, or more specifically *surgical cricothyroidotomy*, is a surgical technique for placing an airway through the cricothyroid space (see Fig. 40-1). The term has also been applied to percutaneous placement of a cannula by Seldinger technique (*cannula* or *percutaneous cricothyroidotomy*) or a needle (*needle cricothyroidotomy*) through the cricothyroid space to establish an emergency airway. Because of its simplicity and the superficial location of the cricothyroid space, cricothyroidotomy has become the preferred procedure for emergency airway placement for patients who cannot undergo translaryngeal intubation.^{125–128} After training on simulators or cadavers, operators can complete surgical or cannula cricothyroidotomy in less than 60 seconds.^{129–131} Some centers, however, prefer emergency PDT after failed intubation attempts.^{21–27} Up to 62% of patients have vertically oriented arteries and veins overlying the cricothyroid membrane, which can complicate cricothyroidotomy.¹³²

Most intensivists do not employ cricothyroidotomy for elective, long-term airway access in critically ill patients because of concern for delayed airway damage, which has been reported to occur more commonly in patients with 6 to 7 days of prior translaryngeal intubation.^{133–137} More recent studies propose that cricothyroidotomy has a low complication rate and represents a reasonable option for critically ill patients with challenging neck anatomy.^{106,138,139} These studies were retrospective, however, and did not provide systematic patient follow-up to detect long-term airway complications.

Cricothyroidotomy presents potential long-term risks to vocalization after decannulation because the tube inserts through the cricothyroid membrane, which lies within 1 cm of the true vocal cords, and can thereby cause glottic and direct vocal cord scarring and also prevent anterior pivoting of the laryngeal cartilage, which is required to stretch the vocal cords.^{140,141} Up to 78% of patients treated with a cricothyroidotomy experience hoarseness.^{140–144} No studies, however, have compared the relative risk of voice changes after cricothyroidotomy as compared with tracheotomy,¹²⁵ which have been reported to cause vocalization problems in 24% of cardiovascular surgery patients after tracheotomy.¹⁴⁵

Because of concerns for long-term airway damage, traditional recommendations propose converting emergency cricothyroidotomies to tracheostomies within 72 hours for patients who require continued airway support.¹⁴⁶ Such recommendations, however, have not been studied. Small case series report serious complications in approximately 50% of ventilator-dependent trauma patients undergoing conversion and longer hospitalizations as compared with patients maintained with their initial cricothyroidotomies.^{18,147} A meta-analysis of 1134 patients undergoing emergency cricothyroidotomy reports a 2.2% incidence of subglottic stenosis overall and a 1.1% incidence in trauma patients, with only 0.3% of trauma patients requiring reconstructive airway surgery.¹²⁵ The study emphasized that the primary studies had considerable deficiencies, no studies support routine conversion, and a need exists for further investigations in light of the potential risks of conversion to tracheotomy.

Although cricothyroidotomy has been recommended for long-term airway support after median sternotomy to avoid mediastinitis, recent reports report an acceptably low risk of mediastinitis when tracheotomy is performed after cardiac surgery.^{148–151} The advent of PDT has further decreased concern regarding contamination of fresh sternal incisions,^{150–153} although PDT remains an independent predictor of deep sternal infections (odds ratio [OR] 3.22, 95% confidence interval [CI] 1.14 to 9.31, $p < 0.0001$) after cardiac surgery.¹⁵⁴

Currently, cricothyroidotomy serves as the primary approach for emergency airway support when translaryngeal intubation fails and as a secondary approach for elective long-term airway access in support of mechanical ventilation. When performed electively, most,^{134,136,142} but not all,^{106,138,147} clinicians reserve cricothyroidotomy for patients intubated less than 7 days. Cricothyroidotomy should also be avoided if possible for patients who depend occupationally on their voice, such as singers or actors,¹⁴⁴ despite absence of comparative outcome data with tracheotomy.¹²⁵

Minitracheotomy

A *minitracheotomy* is a percutaneous technique first described by Matthews et al in 1984 for inserting through the cricothyroid membrane a specialized 4-mm uncuffed tube that accommodates a 10 Fr suction catheter (see Fig. 40-1).^{155–157} Performed at the patient's bedside, minitracheotomy provides direct access to the airway for suctioning tracheal secretions without interfering with the patient's cough or speech. The catheter can be capped when not in use for airway suctioning. Because of the tube's small caliber and uncuffed design, a minitracheotomy does not provide airway access for mechanical ventilation, although it has been used during elective otolaryngologic procedures,¹⁵⁸ in patients undergoing jet ventilation,¹⁵⁹ for patients with sternal dehiscence,¹⁶⁰ and for the management of sleep apnea.¹⁶¹

The procedure may provide benefit for patients with adequate spontaneous respirations who appear at risk for secretion-related deterioration of lung function.^{157,162–168} Patients at high risk for pulmonary complications after thoracic or upper abdominal surgery may benefit from the prophylactic placement of a minitracheotomy at the end of the operative procedure.¹⁶² It has also been used as a transition from PDT for patients who have weaned from mechanical ventilation but who cannot manage secretions.¹⁶⁹ Little outcome data exist, however, to demonstrate efficacy of minitracheotomy with a recent meta-analysis of primary studies demonstrating no benefit in terms of mortality or ICU length of stay for high-risk patients undergoing thoracotomy and lung resection.¹⁷⁰

Minitracheotomy is usually well tolerated, although 1% develop life-threatening complications¹⁷⁰ and 6% to 57% develop minor complications, such as temporary discomfort, voice changes, subcutaneous emphysema, bleeding, and dyspnea.^{156,170–173} Life-threatening complications include pneumothorax after misplacement into paratracheal tissue, profuse bleeding from anterior jugular veins, and esophageal puncture.^{174–177} Long-term studies have not reported subglottic stenosis after minitracheotomy.^{2,157,163}

Surgical Techniques for Tracheal Cannulation

Open Surgical Tracheotomy

Elective standard tracheotomy is performed in an operating room under general anesthesia or an ICU for patients if adequate lighting, personnel, and equipment can provide the resources available in an operating room. The patient's neck is hyperextended with a rolled towel placed between the shoulder blades to bring the trachea and larynx into a superficial and elevated position. The surgeon identifies anatomic landmarks by palpating the cricoid cartilage, tracheal rings, and thyroid cartilage. Some surgeons perform ultrasound to identify structures.¹⁷⁸

Unless the patient has a vertical scar from a previous tracheotomy, most surgeons perform a 2- to 3-cm transverse incision 2 cm above the suprasternal notch midway between the sternal notch and thyroid cartilage over the second tracheal ring with subsequent dissection of the subcutaneous tissues and platysma. A vertical incision then separates the sternohyoid and sternothyroid muscles from the midline allowing retraction of the strap muscles to reveal the thyroid isthmus. The thyroid isthmus can be pulled superiorly or inferiorly out of the surgical field, but may need to be divided and oversewn to allow visualization of tracheal rings. Vessels can bleed substantially and require electrocautery or suture ligation. Although the anterior jugular veins lie lateral to the incisional plane, communicating venous branches may need to be divided.

The trachea is elevated and stabilized by placing a cricoid hook under the cricoid cartilage or first tracheal ring or using lateral traction sutures around the third or fourth tracheal rings. A vertical incision between the second and third or third and fourth tracheal rings is made with a scalpel, avoiding electrocautery because of flash fire risks. A tracheal wall flap (Björk flap) attached inferiorly to the anterior tracheal wall can be created or a section of the tracheal wall removed. The translaryngeal endotracheal tube is pulled back under direct vision above the tracheal incision. Some surgeons pass the endotracheal tube further into the airway, positioning the tip by bronchoscopic guidance just above the carina, which allows the endotracheal tube cuff to block incisional blood from entering the lower airway.¹⁷⁹

After positioning the endotracheal tube, the tracheal incision is gently dilated laterally sufficiently to allow insertion of a tracheotomy tube. The tube size is selected by airway inspection, choosing a tube approximately two-thirds the diameter of the tracheal lumen at the level of the stoma. Some surgeons measure the depth of the stoma track and the angle between the stoma and trachea in all patients to determine a need for a nonstandard tube.¹⁸⁰ Specialized tubes are usually required for patients with obesity, short necks, or other anatomic variations,⁷⁸ with special attention directed toward assessing the depth of pretracheal tissue before tube insertion.¹⁸¹ The nonstandard internal diameters of tracheotomy tubes from different manufacturers should also be recognized.¹⁸² After inflation of the tube cuff and confirmation of adequate ventilation, the endotracheal tube is removed. Some surgeons use bronchoscopy to confirm correct tracheotomy tube positioning and to suction bloody secretions.¹⁸³ The tracheal traction sutures are removed because they serve little value in facilitating recannulation of the trachea if inadvertent decannulation later occurs.

Percutaneous Dilational Tracheotomy

The PDT Blue Rhino single-dilator Seldinger technique utilizes a kit that contains a J-wire guide, Teflon catheter with introducer needle, a Teflon introducer dilator, a translucent Teflon guiding catheter, and a single curved dilator. Some operators first examine the neck after the application of positive end-expiratory pressure with the patient in the horizontal position and the neck slightly extended to detect aberrant jugular veins that traverse the operative field.¹⁶⁹ Others examine the neck by ultrasonography to both detect aberrant vessels and select an insertion site.^{178,184} The patient is preoxygenated with 100% O₂ and positioned with the head extended as for a standard surgical

tracheotomy.¹⁸⁵ Patients with mild to moderate respiratory acidosis may benefit from increasing the minute ventilation before the procedure.¹⁸⁵ Even with a normal baseline partial pressure of carbon dioxide (P_{CO_2}), 26% of patients experience an increased P_{CO_2} by equal to or greater than 3 to 4 mm Hg during insertion of a bronchoscope, with risks for intracranial hypertension in neurosurgical patients¹⁸⁵ unless minute ventilation is adjusted upward.¹⁸⁶ A 2-cm transverse skin incision is made over the second tracheal interspace. Protrusion of fat into the wound is a reliable measure of adequate depth of the incision.¹⁸⁵ Blunt dissection through the incision identifies the anterior tracheal wall.

A fiber-optic video bronchoscope is inserted through the endotracheal tube to visualize the puncture site and ensure its midline position, orientate the endotracheal tube above the needle insertion site, provide suction for pulmonary toilet, and avoid puncturing the posterior tracheal wall.^{111,187,188} After withdrawal of the endotracheal tube above the first tracheal ring and reinflation of the cuff, a syringe with the catheter-introducer needle is advanced between the first and second or second and third tracheal rings. Some operators use the bronchoscope to transilluminate the needle insertion site or depress the insertion site with mosquito forceps under bronchoscopic visualization to ensure precise needle insertion¹⁸⁹ and adequate withdrawal of the endotracheal tube above the needle insertion site.^{187,190} After air is aspirated through the needle, the J-wire is passed into the trachea toward the carina and the needle is removed. The wire must remain freely mobile throughout the procedure or perforation of the posterior tracheal wall should be suspected.¹⁸⁵ The curved dilator is passed over the wire until the trachea has been dilated sufficiently to receive the tracheotomy tube, which is loaded onto an introducer and inserted over the guidewire. Dilation of the trachea may be difficult in young patients with healthy tracheal tissue.¹⁹¹

Some physicians employ methods other than bronchoscopy to ensure proper placement of the tracheotomy tube. A “modified PDT” employs direct palpation of the trachea without bronchoscopy.¹⁹² This technique, however, produces a larger stoma similar to a standard surgical tracheotomy and obviates the advantages from PDT of a snug stoma.¹⁸⁹ Others use an external laser light source for transillumination to obviate bronchoscopy.¹⁹³ Intratracheal placement of the needle and dilator guided by capnographic monitoring of exhaled CO_2 has been reported in a randomized trial of fifty-five patients to have similar outcomes as with bronchoscopy.¹⁹⁴ Insertion of a light wand into the trachea can also guide PDT by transilluminating pretracheal tissue.¹⁹⁵

Most operators, however, use bronchoscopy to guide PDT because of its multiple advantages.^{35,123,196} Bronchoscopy ensures withdrawal of the endotracheal tube above the surgical site,¹⁹⁷ decreases risk of injury to the posterior membranous portion of the trachea,^{35,198,199} allows early recognition of tracheal puncture if it occurs,¹⁹⁹ and may lower risk of pneumothorax and pneumomediastinum.³⁵ Bronchoscopy facilitates translaryngeal reintubation if the endotracheal tube inadvertently moves above the glottis.¹⁸⁵ The endotracheal tube size must be 7.5 mm or greater to accommodate the bronchoscope.¹⁸⁵

Preoperative assessment with ultrasonography may ensure proper placement of a PDT, avoidance of vascular structures, and detection of a deep lying trachea that would indicate a need for surgical tracheotomy.^{200,201} One study in seventy-two patients found that ultrasound imaging altered the originally selected needle insertion site in 24%

of patients because overlying vascular structures and/or thyroid tissue.²⁰² Ultrasonography also may be of value in patients with obesity, short necks, and other anatomical variations.⁸² Ultrasonography may be increasingly used as an adjunct to bronchoscopic PDT with the advent of highly portable ultrasonographic equipment.²⁰³

The airway becomes unstable during PDT when the endotracheal tube cuff is withdrawn into the endolarynx. An additional variation of technique employs removal of the endotracheal tube and insertion of a laryngeal mask airway,^{204–207} microlaryngeal tube,²⁰⁸ or 4-mm pediatric endotracheal tube.²⁰⁹

Cricothyroidotomy

The cricothyroid membrane is located by palpation of the prominence of the cricoid cartilage. The membrane is 9 to 10 mm in height and trapezoidal in shape, with a surface area of 3 cm². It lies 9 to 10 mm beneath the true vocal cords.^{126,210}

To perform a surgical cricothyroidotomy, the patient is positioned as for a standard tracheotomy.^{127,128} A horizontal 2-cm skin incision is carried through the subcutaneous tissue to the thyroid cartilage. In the emergency setting, a vertical skin incision avoids severing the anterior jugular veins.⁷⁸ The lower border of the cricothyroid membrane is incised transversely, and a tracheal hook is placed under the thyroid cartilage. A Trousseau dilator is inserted through the membrane with gentle vertical dilation to allow passage of a 6- or 7-mm tube. The outside diameter of the tube is limited by the height of the cricothyroid membrane (usually 9 mm).

Percutaneous cricothyroidotomy avoids a surgical incision²¹¹ and represents the preferred route in some algorithms for managing difficult airways.²¹² Several prepackaged kits are available (Quicktrach Kit, Teleflex Medical, Research Triangle, NC; Portex Minitrach II Kit, Smiths Medical, Dublin, OH; Melker Kit, Cook Critical Care, Bloomington, IN). The various techniques for percutaneous cricothyroidotomy, necessary training for competency,²¹³ application of ultrasound guidance,^{203,211} and comparative outcomes^{130,211,214,214–216} are discussed elsewhere.

Minitracheotomy

Minitracheotomy can be performed either by a scalpel or Seldinger technique.^{156,164} A scalpel kit has a blade that protrudes 1.4 cm from a plastic guard that limits the depth of penetration through the cricothyroid membrane. After positioning as for standard tracheotomy, the cricothyroid membrane is located and marked by palpating the cricoid cartilage. After tissue infiltration with local anesthetic, the guarded scalpel is inserted through the midline of the cricothyroid membrane into the trachea to produce a 1-cm stab incision. The lubricated curved introducer is passed through the incision. A minitracheotomy cannula is then passed over the introducer, which is then removed from the airway.

The Seldinger technique uses a 16-gauge needle that is passed through the cricothyroid membrane. After confirmation of correct placement by aspiration of tracheal air, the needle is angled caudally and a guidewire inserted. The needle is then removed and a minitracheotomy tube loaded over a vein dilator is passed over the guidewire. Once in place, the minitracheotomy tube should be plugged when not in use for suctioning to prevent the inhalation of dry ambient air.

Complications of Tracheotomy

Techniques and tube designs are sufficiently advanced to allow the safe application of tracheotomy in most ventilator-dependent patients. Although 9% to 40% of patients will experience some type of complication, most of these are minor and the mortality rate is less than 1% (Table 40-1).^{17,217–220} The frequency and seriousness of complications correlate with an institution’s and operator’s expertise in airway management and the use of strict management protocols.^{220,221}

Table 40-1: Complications of Tracheotomy

Aspiration pneumonia
Cardiopulmonary arrest during the procedure
Herniation and fracture of tracheal rings
Inadvertent decannulation
Mediastinitis
Peristomal cellulitis
Pneumomediastinum and pneumothorax
Poor stoma healing after decannulation with scar, keloid, or tracheocutaneous fistula
Posterior tracheal wall perforation
Stomal erosion or breakdown
Stoma site infection
Stomal hemorrhage
Subglottic stenosis or atresia
Surgical emphysema
Tracheal dilation
Tracheal granulomas with obstruction
Tracheal ring rupture
Tracheal stenosis
Tracheoesophageal fistula
Tracheoinnominate fistula
Tracheomalacia

Intraoperative Complications

Cardiorespiratory Arrest

Sudden cardiorespiratory arrest is the most feared complication of tracheotomy, occurring in fewer than 1% of patients. Underlying etiologies include vasovagal reactions, misplacement of the tracheotomy tube, tension pneumothorax, arrhythmias, and pulmonary edema after relief of transient upper airway obstruction.^{35,222,223}

Hemorrhage

Major hemorrhage occurs rarely as an early complication because it can be avoided by identifying anterior jugular veins, vascular anomalies, and the thyroid isthmus to avoid inadvertent transection. Tracheotomies performed below

the fourth tracheal ring risk injury to the innominate artery.

Minor hemorrhage occurs in 1% to 40% of surgical tracheotomies, with bleeding rates depending on operator experience and presence of bleeding diatheses.⁹⁹ Recent PDT series report less than 2% to 4% bleeding rates,^{42,76} although bleeding is the most common individual complication of the procedure.^{35,76}

Pneumothorax and Pneumomediastinum

Pneumothorax and/or pneumomediastinum occur in 0% to 4% of patients,^{97,99,106,119,224} as a result of dissection of air through the incision into the mediastinum, rupture of a lung bleb if transient airway obstruction occurs, or direct injury to the apical pleura.²²⁵ Massive mediastinal insufflation can result from tube misplacement into paratracheal tissue.^{226,227}

Recurrent Laryngeal Nerve Injury

The recurrent laryngeal nerves lie along the tracheoesophageal recesses. A properly placed midline surgical tracheotomy incision should not injure the nerves in their deep positions, but risk of injury is increased in patients with altered cervical anatomy.

Tracheoesophageal Fistula

Improper technique can puncture or lacerate the posterior membranous tracheal wall and cause an acute tracheoesophageal fistula. Most contemporary studies report an incidence rate of 0% to 1% of tracheal perforation during PDT performed with bronchoscopic guidance.^{198,217,219,220,228,229}

Early Postoperative Complications

Hemorrhage

Wound hemorrhage may first present during the early postoperative period when an injured blood vessel ruptures when the patient coughs or moves. Resolution of intraoperative hypotension or dissipation of tissue epinephrine infiltrated during tracheotomy may also cause bleeding to present after the patient leaves the operating room.

Prolonged oozing that persists for longer than 2 to 3 days is usually attributable to coagulopathy.⁹³ Onset of hemorrhage 48 hours or more after surgery suggests a tracheoinnominate fistula.²³⁰ Immediate airway hemorrhage may be less common with PDT as compared with surgical tracheotomy, with rates less than 2%,⁴² although instances of massive, fatal airway hemorrhage with PDT have been reported.^{231,232}

Subcutaneous Emphysema

Subcutaneous emphysema occurs in less than 10% of patients undergoing surgical tracheotomy.^{224,233} Positive pressure escapes from the airway around an inadequately sealed tracheotomy tube cuff and decompresses into cervical tissue planes. Avoiding gauze packing in the tracheotomy wound decreases the risk of subcutaneous emphysema. Use of a fenestrated tube during the first week after tracheotomy promotes subcutaneous emphysema because the fenestrations may shift into the stoma tissue tract.²³⁴ Misplacement of a tube into paratracheal tissue may

first present with subcutaneous emphysema.⁶⁸ Pneumoperitoneum may also occur after tracheotomy.²³⁵ After correction of the underlying cause, subcutaneous air is usually resorbed spontaneously.

Inadvertent Decannulation

Inadvertent decannulation is a life-threatening complication, particularly during the first 72 hours after tracheotomy. During this period, the stoma tract has not fully developed and parastomal tissue can obscure the tracheal window during recannulation efforts. Blind attempts to replace the tracheotomy tube usually result in misplacement into the pretracheal fascia and external airway compression.

Ventilator-dependent patients who experience early inadvertent decannulation should be reintubated through the translaryngeal route if upper airway obstruction is not a factor. The tracheotomy tube can then be reinserted under more controlled conditions. If emergent recannulation of the stoma tract is necessary because of upper airway obstruction, the patient should be positioned as for surgical tracheotomy with a hyperextended neck. Pulling on tracheal traction sutures may allow visualization of the tracheal window. Initial insertion of a smaller size cannula or placement of a guide catheter, such as an intubation stylet, over which a tracheotomy tube is passed, may assist recannulation.

If the tracheal lumen is not clearly seen, a pediatric laryngoscope may assist exploration of the wound and airway visualization. In thick-necked individuals, initial placement of a cuffless pediatric translaryngeal endotracheal tube followed by recannulation with a larger tracheotomy tube later may be required. A fiber-optic bronchoscope or laryngoscope may serve as a guiding stylet to assist tracheotomy tube reinsertion. Regardless of the approach, recannulation should be attempted only by skilled and adequately prepared operators because of the high risk of misplacement.

The 1% to 7% incidence of early inadvertent decannulation can be decreased by appropriately securing the tracheotomy tube.⁹⁹ Tracheotomy tape should wrap closely around the neck, allowing sufficient space for insertion of a single finger. Tape should not be secured over gauze dressings that may later shift. Although avoided by some surgeons, suturing the tracheotomy plate to the skin is an additional preventative approach. Risks of accidental decannulation are especially high in obese patients, who benefit from selection of an appropriately configured tube to fit through the stoma tract.¹⁸¹

Stoma Wound Infection

Although rapidly contaminated with nosocomial pathogens,²³⁶ only 11% to 30% of surgical tracheostomies^{17,111,237} and 4% to 9% of PDT procedures^{111,237} are complicated by wound infections, probably because the stoma is left open to drain secretions. Patients with purulent wound drainage without tissue infection usually respond to local tracheostoma care. Systemic antibiotic therapy is reserved for parastomal cellulitis and signs of deep tissue infection. Necrotizing stomal infections can dissect into cartilaginous tracheal structures, adjacent major blood vessels, and the mediastinum.²³⁸ This complication requires drainage, débridement, and replacement of the tracheotomy tube with a translaryngeal endotracheal tube.²³⁹

Pneumonia

All ventilator-dependent patients are at risk for ventilator-associated pneumonia (VAP). Tracheotomy has theoretic potential for both increasing the rate of VAP by promoting drainage of colonized stomal pathogens into the lungs and for decreasing VAP rates by allowing removal of the endotracheal tube and promoting normal glottis closure, which may prevent aspiration. However, up to 35% of patients with a tracheotomy experience aspiration,^{240–242} which is most often silent.²⁴² No evidence exists that aspiration occurs more commonly in ventilator-dependent patients after conversion to a tracheotomy.²⁴³

Although observational studies indicate that VAP occurs more commonly in patients with a tracheotomy as compared with patients intubated through the translaryngeal route,^{244–246} it is difficult to control these studies for duration and severity of respiratory failure.²⁴⁷ One study examined risk factors for VAP with multivariate analysis and reported tracheotomy as an independent predictor of VAP (adjusted OR = 3.56).²⁴⁸ Studies also report an increased rate of VAP in patients undergoing tracheotomy while receiving sedation²⁴⁹ or after placement of an internal jugular central venous catheter.²⁵⁰ A retrospective case-control study, however, reported tracheotomy as an independent predictor of a lower rate (episodes per 1000 mechanical ventilator days) of VAP.²⁵¹ The relationship between tracheotomy and risk for VAP appears to be highly complex, with many interactions with other clinical factors, which prevent strong conclusions regarding whether tracheotomy is associated with a higher or lower incidence of VAP in patients undergoing mechanical ventilation.

Tube Obstruction

Partial obstruction of tracheotomy tubes decreases airflow and increases work of breathing for patients receiving partial assist modes of mechanical ventilation (intermittent mechanical ventilation or pressure support). Obstruction usually results from inspissated secretions or clotted blood, but may occur with incorrect selection of cannula size that places the tip of the tracheotomy tube against the tracheal wall or carina. Tracheotomy tubes with a removable inner cannula decrease the incidence of obstruction by secretions.²⁵² Tubes with adjustable flanges can be malpositioned in the airway resulting in obstruction of the lumen.²²² Obstruction can also occur when the tip of the tube causes injury to the tracheal wall with invagination of edematous or granulation tissue into the tube lumen^{188,253} or when patient repositioning abuts the tube against the tracheal wall.²⁵⁴

Late Complications

Up to 65% of patients undergoing tracheotomy in the ICU experience some type of late complication, which may correlate with the duration of preexisting translaryngeal intubation.²⁵⁵ The seriousness of these complications requires physicians to monitor patients for complications before and after decannulation.

Poor Airway Alignment

The short length and standardized design of tracheotomy tubes combined with anatomical variation of patients' airways promote malpositioning of the tube tip within the tracheal lumen. Malpositioning increases airflow resistance and ventilatory workload contributing to failure of weaning from mechanical ventilation. One study of mechanically ventilated patients with a tracheotomy at an acute care weaning center reported that 10% of patients had more than

50% occlusion of tube lumens by tracheal tissue, which was associated with prolonged weaning.²⁵⁶ Replacing a tracheotomy tube with one of different design can improve alignment within the trachea.

Tracheal and Laryngotracheal Stenosis

Tracheal stenosis occurs at the level of the cuff, tracheostoma site, or tip of the tube. Low pressure–high volume cuffs lower, but do not eliminate, risk of tracheal stenosis at the cuff site.^{257,258} Inappropriate cuff overinflation or use of too small of a tracheotomy tube that requires cuff overinflation to maintain an airway seal can convert a low-pressure to a high-pressure cuff system. High intracuff pressures transmit to the tracheal mucosa as high cuff tensions that generate pressure necrosis. Resultant mucosal ulcerations become confluent and expose cartilaginous rings that become infected, which leads to weakening of the anterior and lateral tracheal walls.²⁵⁵ Necrosis generates fibrous scars and transmural airway narrowing. Nontransmural tracheal stenosis can also develop when cartilaginous structures remain undamaged but granulation tissue and proliferative scars in a weblike pattern narrow the airway. The greatest area of scar formation is usually within 3.5 cm of the stoma and ranges from 0.5 to 4 cm in length.

Tracheal stenosis can occur at the distal tip of the tracheotomy tube if a malpositioned tube abuts tracheal mucosa. A tracheotomy tube design should be selected that ensures collinearity with the tracheal lumen, and traction on the tube or ventilator hoses should be avoided.

The stoma has become the most common site for tracheal stenosis after surgical tracheotomy and PDT.^{114,122,219,258–262} The airway is narrowed in an anterolateral dimension with relative preservation of the posterior wall. Contributing factors include an overly large tracheal incision and excessive movement of the tube against the tracheal stoma. Fracture of cartilaginous rings during PDT also leads to tracheal stenosis. Patient-related risk factors include female gender, obesity, diabetes mellitus, hypertension, cardiovascular disease, and current smoking.²⁵⁸

Tracheotomy can also cause laryngotracheal stenosis above the stoma site.^{259,263} One observational study reported a higher incidence of suprastomal stenosis with PDT (24% of patients with more than 50% of lumen) as compared with surgical tracheotomy (7% of patients).²⁶⁴ A study in cadavers demonstrated the potential of PDT to damage the cricoid cartilage and structures above the stoma site, which may promote subglottic stenosis.²⁶⁵ A meta-analysis that compared complications from PDT and surgical tracheotomy, however, found no differences in risk of subglottic stenosis.¹²²

Clinical manifestations of tracheal stenosis may develop while the tracheotomy tube is in place, but more commonly occur 2 to 6 weeks after decannulation. Delayed onset as late as 4 months after extubation may also occur. Patients with compromised pulmonary function or neuromuscular disease may not manifest symptoms of tracheal stenosis until an episode of bronchitis with increased airway secretions occurs.

The symptoms and signs of tracheal stenosis may be obscured by the patient's underlying disease, but typically include difficulty clearing secretions, dyspnea unresponsive to bronchodilators, exertion-related dyspnea, cough, monophasic wheezing, stridor, and pneumonia. Patients with normal underlying lung function may be relatively asymptomatic until the tracheal lumen is decreased by more than 50%. Clinically important tracheal stenosis occurs in 1% to 11% of patients after 1 year of follow-up.^{59,114,219} Because of the nonspecific nature of the clinical manifestations of tracheal

stenosis, patients with deteriorating respiratory function and a history of previous intubation should undergo evaluation.

The diagnostic evaluation relies on imaging and endoscopic studies to detect airway narrowing, which may be fixed or dynamic.²⁶⁶ Chest radiographs and standard chest computed tomography (CT) scans have limited sensitivity, but three-dimensional spiral CT reconstruction imaging provides a valuable adjunct.^{267,268} Pulmonary function tests are insensitive (require 80% narrowing) and nonspecific indicators of upper airway obstruction and provide limited diagnostic utility.²⁶⁹ Direct bronchoscopic visualization remains the gold standard. Most patients with airway stenosis are candidates for corrective interventions either by tracheal resection and reconstruction, neodymium:yttrium-aluminum-garnet (Nd:YAG) laser therapy, mechanical or balloon dilation, or placement of a stent.^{270,270–273}

Tracheomalacia

Extensive destruction and necrosis of tracheal cartilages by the tracheotomy tube and ensuing infection of deep tracheal tissue causes tracheomalacia. This complication promotes expiratory tracheal collapse, which can present before or after decannulation. This complication should be suspected in patients who require overdistension of the tracheotomy cuff to maintain an adequate seal.²⁷⁴ Patients may require a longer tracheotomy tube, bronchoscopic stenting, surgical resection, or tracheoplasty.^{183,273}

Tracheoesophageal Fistula

A tracheoesophageal fistula occurs in less than 1% of patients and results from pressure necrosis of the tracheal and esophageal mucosa caused by the tube cuff.^{43,275–277} Risk factors include high cuff pressures, high ventilator-induced airway pressures, excessive tube movement, prolonged intubation, presence of a nasogastric tube, and diabetes mellitus.^{276,278} During the course of intubation or after extubation, patients demonstrate increased cough and/or tracheal secretions. Symptoms may only occur after eating solids or swallowing fluids. Patients undergoing mechanical ventilation may experience gastric distension or frequent belching, and chest radiographs may show an air-filled esophagus or gastric distension. Recurrent aspiration may manifest as pneumonia.

A high index of suspicion for tracheoesophageal fistula is required for critically ill patients who have compatible symptoms that may mimic other swallowing or pulmonary problems. The diagnosis is made by tracheoscopy and esophagoscopy, which determine the location and extent of the fistula. Positioning the cuff of an adjustable tracheotomy tube distal to the fistula (Fig. 40-3) and placement of a gastric tube limit airway soiling by aspirated gastric contents. A jejunostomy tube provides nutritional support. Because spontaneous closure rarely occurs, surgical correction is required, which must be delayed until after weaning from mechanical ventilation to avoid anastomotic dehiscence. Primary closure can correct small tracheal fistulae, but most patients require tracheal resection and reconstruction.^{276,279} Tracheoesophageal fistulae are universally fatal if left uncorrected.

Figure 40-3



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Flexible tracheotomy tubes with adjustable flanges allow customization of a tube for fitting patients with difficult anatomy or a special need to place the cuff over a specific segment of the airway. (Courtesy TRACOE Medical, Frankfurt, Germany.)

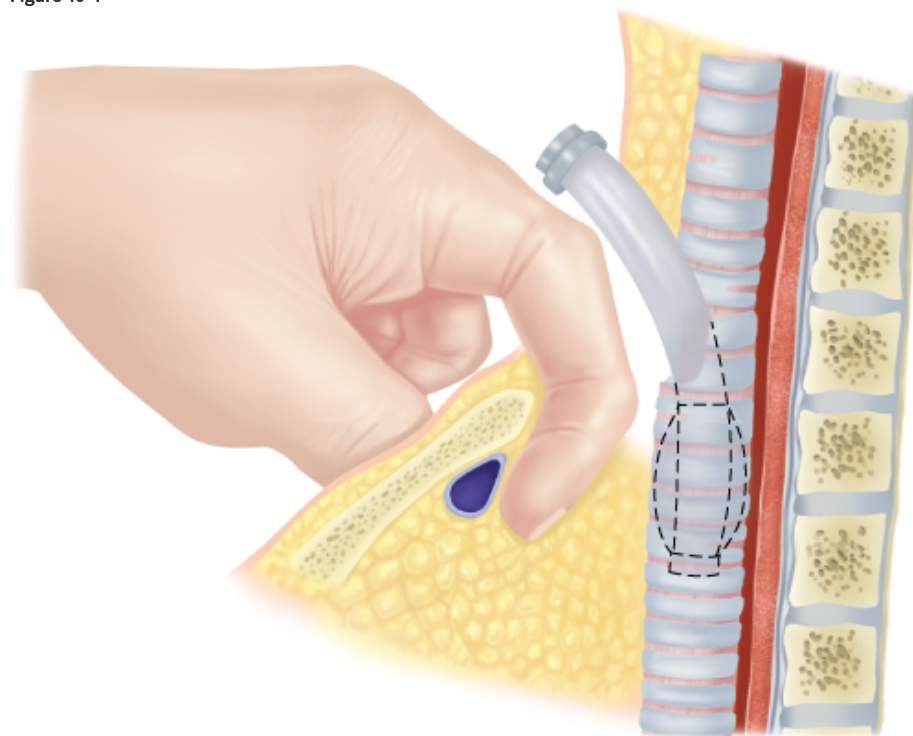
Tracheoarterial Fistula

A tracheoarterial fistula is a major, potentially life-threatening complication of tracheotomy that occurs in less than 1% of patients.^{280–283} The innominate artery is most commonly involved because it is nine to twelve tracheal rings below the cricoid cartilage within reach of the tip, and sometimes the cuff, of the tracheotomy tube. In some patients, the innominate artery can traverse the midline just below the tracheotomy stoma where the “elbow” of the tube just above

the cuff can induce pressure necrosis.²⁸⁴ Arterial erosions, however, may also involve the common carotid artery, inferior thyroid artery, the thyroid ima artery, the innominate vein, and the aortic arch.²⁸⁵ Improper traction on the tracheotomy tube by ventilator tubing, placement of the tracheal window below the fourth tracheal ring, misalignment of the tube with the tip against the tracheal mucosa, high cuff pressures, prolonged intubation, excessive patient movement or posturing, sepsis, malnutrition, and corticosteroid therapy increase the risks for this complication.^{286–289}

Massive airway hemorrhage and asphyxia are the feared complications of tracheoarterial fistula. Hemorrhage can occur as early as several days or as late as 7 months after tracheotomy with the peak incidence between the first and third weeks.^{280,283,286,290,291} Up to 50% of patients who experience an episode of airway hemorrhage later than 72 hours after a tracheotomy have an underlying tracheoarterial fistula.²⁸⁶ Pulsations of the tracheotomy tube and “spotty” herald hemorrhages may portend massive hemorrhage. Any airway bleeding 72 hours after tracheotomy warrants consideration of fiberoptic endoscopy in an operating room environment in case airway manipulation precipitates massive hemorrhage.^{230,280} Three-dimensional CT scanning may establish the diagnosis for stable patients.²⁹² Patients who present with massive hemoptysis should be managed with overinflation of the tracheotomy tube cuff or insertion of a translaryngeal tube with positioning of the cuff over the fistula in an effort to tamponade bleeding. A finger inserted into the stoma tract may allow compression of the innominate artery anteriorly against the sternum (Fig. 40-4).^{286,287}

Figure 40-4



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Hand position for finger tamponade of the innominate artery in a patient with a bleeding tracheoinnominate fistula.

A tracheoarterial fistula represents a surgical emergency and patients require a median sternotomy with ligation and resection of the artery because tissue infection obviates vascular repair.^{280,283,285,293} Interruption of the innominate

artery is usually tolerated without neurologic sequelae.^{280,285,293}

Tracheocutaneous Fistula and Stomal Scars

Epithelialization of the stoma tract may prevent the stoma from closing spontaneously after decannulation. The parastomal platysma or a sternohyoid muscle flap can then be pulled over a surgical stoma repair followed by skin closure.^{294–296} Some patients develop excessive scarring at a closed stoma site, which can be managed through reconstructive surgery that fills lost deep tissue bulk, corrects any tracheal skin tugging, and produces a tension-free closure that falls more naturally into the neck folds.²⁹⁷

Tracheal Ring Fracture and Herniation

A PDT risks fracture and herniation of tracheal rings because the technique dilates rather than incises intercartilaginous tissue and produces an anteroposterior vector of force at the tracheal insertion site.^{298–303} Damaged rings can protrude into the airway causing tracheal narrowing³⁰⁴ or promote tracheomalacia if extensive loss of cartilaginous support occurs.³⁰⁵ The frequency of clinically important complications after tracheal ring fractures is unknown. One study followed sixteen patients with tracheal ring fractures and found no instances of tracheal stenosis.³⁰⁶

Timing of Tracheotomy during Mechanical Ventilation

Although translaryngeal endotracheal tubes are relatively well tolerated, patients who require prolonged ventilation eventually benefit from tracheotomy (Table 40-2).^{307,308} Selecting patients for tracheotomy and identifying the ideal time for the procedure, however, remain clinical challenges with physicians demonstrating considerable practice variation.^{41,309–311} Clinical trials that randomize patients to “early” versus “late” tracheotomy to inform timing decisions face major difficulties.³¹² Investigators must be able to identify accurately soon after intubation those patients who will most likely require prolonged ventilation so as to avoid performing unnecessary tracheotomies.³¹³ Unfortunately, experts doubt their own abilities to predict prolonged intubation during the first days of ventilation.³¹⁴ Recent trials report that a large proportion of enrolled patients randomized to control, nontracheotomy groups based on their high risk of needing prolonged ventilation actually undergo early extubation.^{315,316} Collected study data should ideally include comprehensive short-term (e.g., pneumonia, incisional bleeding, duration of mechanical ventilation, patient comfort) and long-term (e.g., tracheal stenosis, speech problems, dysphagia) outcomes, which often have nonstandardized definitions, imprecise or costly approaches to detection, and dependency on case mix.

Table 40-2: Advantages of Tracheotomy as Compared with Prolonged Translaryngeal Intubation

Decreased airway resistance for promoting weaning from mechanical ventilation
Earlier transfer of ventilator-dependent patients from the ICU
Enhanced oral nutrition and oral hygiene
Enhanced phonation and communication
Improved airway suctioning
Increased comfort with opportunities to decrease sedation
Increased patient mobility
Less direct endolaryngeal injury
More secure airway and decreased risk of inadvertent decannulation

Additional study design challenges include (a) physician reluctance to allocate patients to the early tracheotomy group when assigned by randomization, (b) difficulties with blinding providers to group allocation, (c) standardizing patient care with protocols for other critical care interventions and approaches to weaning and extubation, and (d) standardizing the tracheotomy technique (i.e., PDT versus surgical tracheotomy). Consequently, most existing prospective trials have limitations because of small and heterogeneous study populations, variations in defining “early” versus “late” tracheotomy, and various methodological flaws that complicate meta-analyses.

These limitations indicate that timing tracheotomy for critically ill patients remains a complex decision that requires individualization of care. No data exist to establish a standard time when tracheotomy must be performed to avoid airway injury related to continued translaryngeal intubation. Moreover, the risks of performing tracheotomy are acceptably low so that it can be safely applied when patients appear likely to benefit from the procedure.

Recommendations to “calendar watch” and avoid tracheotomy if at all possible for at least 14 to 21 days because of its attendant risks have become obsolete. Patients should be selected for tracheotomy by considering the unique circumstances of the patient at hand and the likelihood based on existing studies that the benefits of tracheotomy outweigh its inherent risks as compared with the competing risks of prolonging translaryngeal intubation ([Table 40-3](#)). Patient and family values and preferences should also influence the decision.

Table 40-3: Complications and Disadvantages of Prolonged Translaryngeal Intubation

Airway perforation
Aspiration pneumonia
Cricoid cartilage abscess
Damage to intrinsic laryngeal muscles
Dislocation or subluxation of the arytenoid
Granulation tissue forming interarytenoid adhesion
Healed fibrous glottic nodules
Inadvertent extubation
Mucosal ulceration
Nasal septal fractures with nasotracheal intubation
Nosocomial sinusitis
Otitis media
Oversedation
Patient discomfort
Posterior glottic stenosis
Subglottic stenosis
Tracheal stenosis
Ulcerative lesions of the nares and pharynx
Vocal cord fixation from fibrosis of the cricoarytenoid joint
Vocal cord laceration
Vocal cord paralysis
Subglottic web formation

In considering potential benefits, a tracheotomy provides opportunities to improve a patient's comfort and sense of well-being.^{316–318} For patients receiving intravenous sedation, some,^{319,320} but not all,^{316,321} studies indicate that drug requirements to maintain comfort are decreased after tracheotomy. By freeing the oropharynx, tracheotomy promotes oral hygiene and nutrition and facilitates articulated speech,^{322–325} allows early mobilization,^{312,326} and permits transfer from the ICU to more comfortable and family-centered settings for patients who remain ventilator dependent.³²⁷ More effective oral care and decreased sedative requirements may decrease risks of ventilator-associated pneumonia.^{328,329} The largest available randomized trial that assessed impact of tracheotomy timing on pneumonia risk, however, reported no differences between patients undergoing early versus late tracheotomy.³¹⁵

The shorter length and rigid structure of tracheotomy tubes have been stated to decrease airway resistance and promote weaning from mechanical ventilation.^{330,331} The resistance to airflow of translaryngeal endotracheal tubes in situ increases beyond in vitro predictions because thermolabile tubes assume angulated configurations and become inspissated with secretions, both of which promote turbulent airflow.^{332,333} Studies conflict, however, about whether or not the lower measured resistance of tracheotomy, as compared with endotracheal tubes, produces meaningful benefits. Lin et al noted no clinically important differences in pulmonary mechanics before and after tracheotomy.³³⁴ Three studies, however, noted small decreases in work of breathing after tracheotomy that became more significant

with increasing respiratory rates, in addition to decreases in airway resistance, pressure–time product, and auto–positive end-expiratory pressure.^{335–337} No changes in tidal volume, respiratory rate, or dead space ventilation were observed.^{335–338}

Tracheotomy provides an opportunity to further decrease work of breathing by removing the inner cannula during spontaneous breathing weaning trials.³³⁹ Enhanced ability to suction through a tracheotomy may promote removal of airway secretions, which increase airway resistance of endotracheal tubes; decreased secretions may promote weaning from mechanical ventilation for patients with borderline ventilatory reserve.^{336,340}

Most prospective randomized trials report that early tracheotomy within 7 days of intubation decreases duration of mechanical ventilation.^{4,315,320,327,341–344} Prospective studies of early tracheotomy also decreases ICU stay, possibly by promoting weaning and allowing transfer of ventilated patients with a stable airway to the wards.^{4,315,320,327,344} Some retrospective^{10,345,346} and prospective³⁴⁷ cohort studies indicate that earlier tracheotomy decreases total hospital stay. Other retrospective studies note no differences in hospital length of stay.^{348,349} Of note, however, the largest prospective randomized trial to date of early tracheotomy did not find any effect on duration of hospital stay even though time in the ICU was decreased.³¹⁵ This study, however, excluded patients with chronic obstructive pulmonary disease and pneumonia. If future randomized studies demonstrate tracheotomy does not decrease hospital stay, any decreases in duration of ICU stay would be expected to result in only small total decreases in total costs of hospitalization.³⁵⁰

One randomized trial of early PDT reported decreased hospital mortality for patients in medical ICUs who received early (32% mortality, 2 days before tracheotomy) versus late (62% mortality, more than 14 days before tracheotomy) PDT.³²⁰ The study, however, limited enrollment to patients with an Acute Physiology and Chronic Health Evaluation score of greater than 25, which limits its generalizability to patients with lower mortality risks.³⁵¹ Other prospective trials^{4,315,316,327,341,343,344} and a retrospective review of a large dataset³⁵² demonstrate no mortality benefit. Most retrospective and prospective studies of trauma and burn patients report no benefit on mortality between early and late tracheotomy.^{236,341,343,344,346,348,349,353–357} Guidelines exist, however, that recommend early tracheotomy for trauma patients estimated to require ventilation for more than 7 days.³⁵⁸

Inadvertent extubation occurs in 8.5% to 21% of ventilator-dependent patients^{317,359–361} and exposes patients to risks of adverse cardiopulmonary events that include nosocomial pneumonia.^{360,362–364} Tracheotomy decreases the risk of self-extubation.^{319,320,360,365}

Tracheotomy allows removal of a translaryngeal endotracheal tube and interrupts ongoing injury by the endotracheal tube to intralaryngeal structures. Risks of prolonged translaryngeal intubation include oropharyngeal injury, pressure necrosis of the larynx resulting in subglottic stenosis, and nosocomial sinusitis.^{366–368} The decision to perform tracheotomy would be simplified if evidence existed that the risk of laryngotracheal injury correlated with duration of endotracheal intubation. Aggregate data indicate that some degree of clinically important laryngeal injury occurs in 10% to 19% of patients managed with prolonged translaryngeal intubation.³⁶⁹ It is not clear, however, that laryngeal injury can be predicted by the duration of endotracheal intubation or the detection of acute airway lesions in intubated

patients.^{370–373} No support exists, therefore, that “calendar watching” with the performance of tracheotomy after a specific duration of translaryngeal intubation prevents long-term airway injury.

In contrast to calendar watching, many clinicians endorse “the anticipatory approach” to timing tracheotomy that tailors the decision to perform a tracheotomy to an individual patient’s unique needs.^{308,369,374,375} With this approach, the decision to perform of tracheotomy centers less on “when” tracheotomy should be performed and more on “for whom.” Patients with respiratory failure first receive necessary critical care to allow stabilization. Patients who appear likely to achieve extubation within the first several days of ventilation are not considered for tracheotomy. If patients remain ventilator dependent for several days and appear likely to require ventilation for 7 or more days, a tracheotomy can be considered. The actual decision to perform the procedure, however, depends on multiple patient-dependent factors that determine whether the patient will likely experience the benefits associated with tracheotomy and on patient and family values. The decision can usually be made for most patients within the first week of intubation.³⁷⁶

The anticipatory approach requires daily reassessments of patients to estimate the likelihood that weaning and extubation remains unlikely because the accurate prediction of prolonged ventilation with various scoring systems remains elusive.^{6,315,377–390} A standardized protocol to assess patients for tracheotomy may result in more appropriate application of the procedure and less practice variation.³⁹¹

Special Patient Care Considerations

After tracheotomy, ventilator-dependent patients benefit from a well-organized treatment plan and an interdisciplinary team experienced in tracheotomy management. Skills in evaluating patients for oral communication and nutrition, avoiding airway-related complications, initiating an airway weaning protocol, and recognizing complications after decannulation improve patient outcome.³⁹² These skills are especially important as pressures increase for earlier transfer of patients with tracheotomy tube to wards or intermediate care units.

Management of Tube Cuff Pressure

Tube cuff pressures should be maintained between 18 mm Hg (approximately 24 cm H₂O) to 25 mm Hg (approximately 34 cm H₂O). Cuff pressures greater than 25 mm Hg can tamponade mucosal capillaries and cause mucosal ischemia, leading to tracheal stenosis.³⁹³ Intracuff pressures below 18 mm Hg may promote microaspiration and nosocomial pneumonia.^{394–396} Various techniques exist to estimate cuff pressure, such as finger palpation of the external inflation bulb, minimal occlusive volume, and subjective assessment of tension required on a syringe to inflate the cuff. None, however, reliably substitutes for direct cuff pressure measurements with a pressure gauge.^{397–401} In the absence of routine cuff pressure monitoring, up to 45% of critically ill patients have excessively high endotracheal tube cuff pressures.⁴⁰² Unfortunately, use of manometry to monitor cuff pressure does not ensure the avoidance of excessively high inflation pressures because of measurement practice variation.^{403,404} Chest radiographs can detect overdistension of the tracheal lumen by a hyperinflated cuff and provide an additional safeguard against overinflation.⁴⁰⁵ Tube cuffs from different manufacturers may seal airways at different pressures.⁴⁰⁶

Several factors contribute to elevated cuff pressure. High peak airway pressures externally compress the cuff surface raising intracuff pressure and tracheal wall tension.⁴⁰⁷ Patients undergoing general anesthesia experience a rapid increase in cuff pressure as anesthetic gases infuse into the air-filled cuff.⁴⁰⁸ An inappropriately small tracheotomy tube can also cause increased tracheal wall tension with resultant delayed airway stenosis by necessitating cuff overinflation to maintain an airway seal.⁴⁰⁹ Elevated cuff pressures may signify the presence of tracheal dilation (e.g., tracheomalacia), which requires cuff overdilation to produce a seal.²⁷⁴ Transport to high altitude can also increase cuff pressure.⁴¹⁰

Standards of respiratory care call for recording cuff pressures measured by a calibrated device and recording pressures once a shift or whenever the tube is changed or other events occur that may alter cuff pressure.⁴¹¹ Cuff pressure should be monitored with the use of a syringe, stopcock, and manometer that allow simultaneous communication between all three compartments while pressure is being adjusted and measured.⁴¹¹ Use of the minimal leak technique or palpation of the pilot balloon is not recommended.^{409,411}

Swallowing

Normal swallowing is a complex physiologic event comprised of simultaneous and sequential contractions of muscles of the oral–facial region, pharynx, larynx, and esophagus. To initiate swallowing, upward and backward motion of the tongue moves a food bolus to the posterior oral cavity or oropharynx,^{412–414} where stimulation of neuroreceptors triggers pharyngeal swallowing and halts respiration, typically in the expiratory phase.^{415–418} The pharyngeal swallow comprises five overlapping events that protect the airway and clear the pharynx of ingested material. These events are (a) elevation and retraction of the soft palate, (b) elevation and anterior displacement of the hyoid bone and larynx, (c) laryngeal closure, (d) pharyngeal contraction, and (e) opening of the pharyngoesophageal region.^{412,419–424}

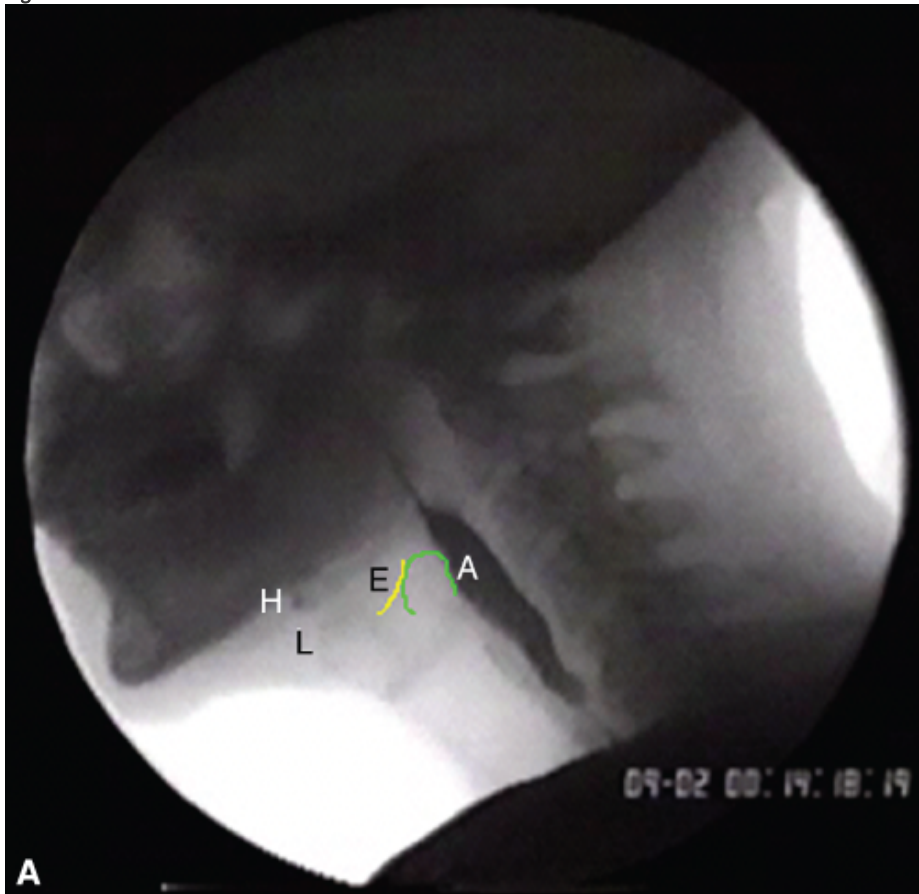
Contraction of the pharyngeal constrictors occurs coincident with forceful retraction of the tongue, which propels the bolus posteriorly to assist pharyngeal clearance.⁴²⁴ Contractions of the thyrohyoid and submental extrinsic tongue muscle group elevate and move forward the hyoid bone and larynx as a functional unit. This hyolaryngeal complex movement closes the larynx to prevent aspiration and pulls the cricoid cartilage anteriorly and away from the posterior pharyngeal wall, thereby opening the pharyngoesophageal segment region to permit entry of the bolus into the cervical esophagus.^{420–426} The pharyngoesophageal segment becomes compliant by synchronized relaxation of the cricopharyngeal muscle. The larynx then descends toward its resting position and respiration resumes characterized by a small expiratory airflow.^{415–417} Once in the cervical esophagus, the bolus is propelled by primary and secondary esophageal peristaltic muscle contractions.^{427,428} These contractions continue until the bolus progresses into the stomach through the passively relaxed lower esophageal sphincter.

A tracheotomy tube and/or inflated cuff may interfere with the mechanics of swallowing (Fig. 40-5). A tube can anchor the hyolaryngeal complex and prevent its normal excursion during swallowing causing incomplete airway closure and penetration of liquid and food into the laryngeal vestibule with aspiration below the vocal folds.^{429,430} Prevention of the normal upward and forward mobility of the hyolaryngeal complex impedes anterior displacement of the cricoid cartilage leading to incomplete opening of the cervical esophagus and retained pharyngeal contents, which may be aspirated into the open airway after completion of the swallow. An inflated tube cuff may disturb esophageal motility

and slow esophageal clearance causing retrograde flow of ingested material into the pharynx with aspiration.

Coexisting gastroesophageal reflux disease with esophageal refluxate increases the risk for aspiration.⁴³¹ Preexisting translaryngeal intubation presents an additional risk factor for aspiration even in patients who do not progress to tracheotomy with more than 50% of patients experiencing dysphagia after prolonged translaryngeal intubation.⁴³²

Figure 40-5



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A. Superior and anterior displacement of the hyoid (*H*) and larynx (*L*) results in approximation of the arytenoid cartilages (*A*) and epiglottic petiole (*E*), and release of the posterior cricoid area from the posterior pharyngeal wall. **B.** The hyoid (*H*) and larynx (*L*) are anchored by the inflated tracheotomy tube cuff leading to incomplete approximation of the arytenoid cartilages (*E*) to the epiglottic petiole (*E*) with resultant penetration, aspiration, and incomplete opening of the cervical esophagus.

Tracheostomized patients at risk for aspiration should undergo a systematic assessment to determine the nature of any swallowing disorder to prevent aspiration and facilitate weaning to decannulation. Coordination of services between speech therapists and critical care providers provides resources for assessing aspiration risk, monitoring the extent of aspiration, and managing risk to prevent complications during tracheotomy use, weaning, and postdecannulation (Table 40-4).^{433–436}

Table 40-4: Clinical Warning Signs Suggestive of Swallowing Impairment and Aspiration

- Adventitious lung sounds
- Articulatory imprecision
- Breathy vocal quality
- Decreased frequency of swallow
- Decreased laryngeal sensation
- Decreased orofacial sensation
- Delayed cough
- Drooling
- Gas-induced cough
- Harsh vocal quality
- Hoarse vocal quality
- Immediate cough
- Impaired resonance
- Labored chewing
- Labored oral transport
- Laryngeal elevation
- Orofacial weakness
- Reduced O₂ saturation
- Sounds of swallow
- Strained vocal quality
- Wet vocal quality

Nutrition

Feeding patients with a tracheotomy through a nasoenteric tube increases the risk of aspiration and pneumonia. Up to 69% of patients with a tracheotomy aspirate at least once every 48 hours regardless of mental status.^{395,437}

Although a tracheotomy offers opportunities to resume an oral diet, patients recovering from critical illness with a tracheotomy in place commonly have severe swallowing dysfunction,^{242,438,439} either from underlying disease or effects of the tube itself.^{440,441} Before resuming oral feeding, patients should undergo evaluation by an interdisciplinary team that includes a speech-language pathologist trained in swallowing assessment.^{433,441,442}

Communication: Voice, Speech, and Language

A tracheotomy disrupts normal speech production. For patients previously intubated with an endotracheal tube, however, a tracheotomy may improve patient well-being by increasing options for communication.^{375,433,443–448} Personal isolation and an inability to communicate during mechanical ventilation are leading factors in increasing anxiety in patients, families, and health care providers.^{324,325,444,448–451}

Several techniques for promoting verbal communication can be tailored to the individual patient's ability to participate in speech therapy. Patients may also graduate through increasingly more complex techniques as medical conditions allow. [Table 40-5](#) outlines the clinical features and applications of these techniques.

Table 40-5: Methods of Communication for Intubated Patients

Communication Option	Advantages	Disadvantages	Considerations	Select Patient Use
Lip reading, mouthing	Follows a natural communication style	Decreased understanding of mouthed messages	Patient should be alert	Can be used with a variety of nonparalyzed patients
	Closely resembles normal speech movement	May result in patient frustration	Requires intact motor functioning	Can be used with patients who are unable to use other alternatives for communication
	Limited training required	Unnatural for patient to decrease rate and length of message	Works best for nasally intubated patients	
	Makes use of oral motor functions	Very limited for orally intubated patients	Requires proper tube taping technique in orally intubated patients for maximal oral movement	
	Allows for natural production of facial expression		Patient at times requires encouragement to mouth words	
	Communication letter board assists communication		Patients often require cueing to decrease rate and length of messages	
Gestural language (yes–no [Y–N] responses, facial expressions, hand gestures)	Y–N responses result in effective and rapid communication of basic needs	Limited interpretation	Establish Y–N response mode and communicate to staff and family (strive for natural Y–N responses with head nods)	Especially helpful for patients with oral intubation and/or oral motor dysfunction

Communication Option	Advantages	Disadvantages	Considerations	Select Patient Use
	• Appropriate facial expressions can greatly enhance communication and emotion	• Difficulty expressing complex or lengthy messages	• Patients may require encouragement to use consistently	
	• Message intent is usually understood	• Requires coordination and dexterity		
Written communication	• Can be used for effective expression of basic needs and wants	• Timely method of communication	• Requires: motor coordination, strength of upper extremity, use of arm free from medical restraints, adequate visual acuity and perception, and supplies at bedside	• Can be used as an alternative to other communication options
		• Requires significant patient endurance and alertness	• May require prepositioning for writing comfort and visibility	
		• Limited expression of complex ideas and emotion	• Use a felt-tip marker or other antigravity dark marking device	
		• Often difficult to read patient writings	• Provide a clipboard for firm writing surface	
		• May require staff assistance	• Provide glasses before writing attempts	
		• Requires interpretation (language, writing style, etc.) by receiver		

Communication Option	Advantages	Disadvantages	Considerations	Select Patient Use
Communication boards (picture, alphabet, and word boards)	• Effective communication of basic needs	• Timely communication process	• Provide glasses to prescription lens wearers	• Can be used with a variety of nonparalyzed patients
		• Communication limited to items on board	• Board should be kept at bedside	• Can be used with patients who are unable to use other alternatives for communication
		• High degree of frustration with alphabet use	• Position board and size of characters to meet patient's needs	
		• Requires coordination for pointing response		
		• Requires intact visual acuity/perception		
		• Potential source of contamination when board used from room to room		
		• Requires immediate accessibility, patient training, and staff patience		

Communication Option	Advantages	Disadvantages	Considerations	Select Patient Use
Augmentative communication devices (“parrot,” “voice-aided,” “text-to-speech” computer software and tablet apps, direct selection, and scanning devices)	• Effective communication of basic needs but newer software (i.e., tablet apps) capable of sophisticated touch screen text-to-speech	• Must be programmed or existing software modified to meet patient needs	• Patient and family need to be included in programming and selection of communication choices	• Can be used with a variety of nonparalyzed patients
	• Provides a programmed speaking voice when text selected or typed	• Communications limited to items on the board or touchscreen unless patient can type	• Provide glasses to prescription lens wearers	• Can be used with patients who are unable to use other alternatives for communication
	• Communication programs can be highly individualized to the patient	• May be frustrating because of limited options or difficulty with typing or use of touchscreen	• Device should be kept at the bedside	
	• Easy access to provide complete phrases	• Requires operator training		
	• Provides a means for more sophisticated, complex communication	• Requires concentration for selection of appropriate phrases		
	• Requires minimal energy once patient is familiar with device	• Requires coordination for pointing response		

Communication Option	Advantages	Disadvantages	Considerations	Select Patient Use
	• Timely communication process	• Potential source of contamination if device is used from room to room		
		• Requires visual acuity		
		• Requires maintenance (software update, battery charging, electrical safety, repairs)		
		• May be costly to the patient (equipment, support of speech therapy)		
Electrolarynx with or without oral adapter and intraoral electronic larynx (multiple brands)	• Provides a voice without using vocal cords	• Poor mechanical quality of voice (monotonal)	• Dentures should be in place	• To be used with tracheotomy patients (free use of mouth for motion)
	• Provides auditory component to speech productions for better understanding	• Requires assistance from staff, significant patient and staff training, dexterity and hand coordination, and intact oral and motor functioning	• Mouth and lips should be moist	
	• Allows verbal communication of basic needs and emotions	• Patient must be alert and cooperative	• Demonstrated technique to patient and family and allow for return demonstration	

Communication Option	Advantages	Disadvantages	Considerations	Select Patient Use
	• Makes use of oral motor functioning	• Patient must have energy reserve to use effectively	• Allow time for experimentation for this and other options (including oral adapter)	
	• Useful when tracheotomy cuff cannot be deflated for other speaking techniques	• Oral adapter may be uncomfortable and awkward		
Deflated/cuff leak methods (cuff leak on ventilator, T-piece)	• Allows use of own voice through vocal cords' natural communication styles	• May be limited to use for short periods depending on tolerance level of patient	• Patients must be alert	• For use with ventilator patients who are stable with cuff leak/deflation and who have complex communication needs
	• Results in efficient communication	• Requires staff assistance to deflate and monitor cuff	• Patients with copious secretions should be suctioned before deflation	• Excellent for chronic/long-term ventilator and tracheotomy patients (especially nonventilator-dependent patients)
	• Requires minimal staff assistance if patient can tolerate deflation	• Methods contraindicated for acute or medically unstable patients	• Procedure must be fully explained to the patient	• Nasotracheal or T-piece patients (stable respiratory status)
	• No cost requirements	• Requires intact oral motor functioning and respiratory reserve	• Ventilator-dependent patients require cueing for proper voice initiation	

Communication Option	Advantages	Disadvantages	Considerations	Select Patient Use
		Requires staff training (to monitor cuff pressures and exhaled volumes)	T-piece users must have T-piece occluded for voice to be audible	
			Ventilator adjustments often needed	
Tracheotomy phonation valves (Olympic, Passy-Muir, Boston Medical)	Allows for phonation and production of patient's own voice	May be limited to use for short periods	Provide support and encouragement during initial valve applications	For use in long-term and chronic tracheotomy patients
	Natural communication method	Requires staff assistance to plug in most patients	Close observation is required	Some may be used for ventilated patients
	Some can be used on ventilated patients	Requires intact oral and motor functioning	Valve is routinely removed when not in use in acute setting	
		Patients without respiratory reserve may not tolerate		
		Works poorly on patients with stiff lungs or with copious secretions		

Patients undergoing positive-pressure ventilation with moderate to low minute ventilation requirements may be able to whisper during brief periods of partial cuff deflation. Patients mechanically ventilated because of neuromuscular disease can be managed by a cuffless or fenestrated tracheotomy tube during positive pressure ventilation and retain the ability to speak.^{452–454} A portion of the ventilator-delivered tidal volume enters the lower trachea and flows around the tracheotomy tube toward the upper airway during inspiration allowing the patient to talk.

The above approach limits speech to inspiration and a short portion of expiration. Speech throughout the respiratory cycle can be promoted by partial cuff deflation as described above, prolongation of the ventilator's inspiratory phase, and application of low levels of positive end-expiratory pressure.^{443,455,456} The resultant continuous air leak around

the cuff allows more spontaneous speech.⁴⁵⁷ Pressure-support ventilation in the presence of a leak around a tracheotomy tube causes an increase in inspiratory time and volume,^{458–460} both of which may theoretically assist speech production.

Specialized tracheotomy tubes, termed *pneumatic speaking tubes* or *talking tubes*, promote speech by supplying pressurized gas mixtures to the trachea above the tube cuff.^{461–464} The gas escapes through the glottis allowing whispered or vocalized speech. A pressurized gas source is connected to a cannula that travels through the wall of the tracheotomy tube and exits at its greater curvature. A gas flow rate of 1.5 to 10 L/min successfully produces intelligible speech in 75% of properly selected patients.⁴⁶⁵ Prognostic indicators for successful use of pneumatic speaking tubes include adequate mental status, availability of a well-trained speech-language pathologist for training and positive reinforcement, absence of laryngeal pathology such as vocal fold paralysis, patient motivation, and encouragement from the critical care staff and patient's family.⁴⁶⁶

Pneumatic speaking tubes may fail because the pressurized gas may flow retrograde through the tracheostoma around the tube. Because this retrograde flow can cause subcutaneous emphysema in patients with recent tracheotomies, pneumatic tubes are contraindicated during the first 5 to 7 days after surgery. Pneumatic tubes may also fail because secretions or granulation tissue above the cuff may occlude the exit port for gas flow.⁴⁶⁷

Other complications of pneumatic speaking tubes include desiccation and inflammation of laryngeal mucosal structures caused by the drying effect of the gas flow. Warming and humidification of the gas source along with intermittent use of the technique can prevent the onset of laryngitis. An air mixture rather than pure oxygen should be used to avoid further airway injury from hyperoxia.⁴⁶⁸

An “electrolarynx” is a valuable device for promoting speech that has a high rate of success in critically ill patients with a tracheotomy.^{469,470} These devices consist of a vibrator unit with an attached handheld battery pack that generates a vibratory tone with variable loudness and pitch. Held firmly against the patient's neck midway between the mandibular angle and the notch of the thyroid cartilage or attached with an intraoral stem that applies intraoral vibrations when inserted through the mouth, the electrolarynx substitutes for the vocal cords as an amplifier for speech. More than 90% of mentally alert patients with a tracheotomy can learn to use an electrolarynx successfully.⁴⁶⁹ Critical care team members may require frequent in-servicing on correct application of the device because of the loss of enthusiasm that develops when patients do not quickly learn the technique.

Patients being weaned from mechanical ventilation who have achieved periods of spontaneous ventilation can speak with the assistance of a fenestrated tracheotomy tube.⁴⁵⁷ Fenestrated tubes with an inner cannula can provide airway access for mechanical ventilation and a system for speech when the inner cannula is removed. Occlusion of the tracheotomy tube with a gloved finger or plug can divert airflow through the upper airway allowing whispered or vocalized speech. Deflation of the tube cuff can further improve expiratory airflow through the larynx.

A fenestrated tracheotomy tube can also be combined with a one-way “phonation valve” (e.g., Passy-Muir valve, Passy and Passy, Irvine, CA) that allows inspiratory airflow through the tube but closes during expiration diverting airflow through the larynx (Fig. 40-6).⁴⁷¹ Phonation valves can be applied to the tracheotomy tube of spontaneously breathing patients and patients undergoing positive pressure ventilation. In the latter setting, it is essential that the patient is being ventilated with the tracheotomy cuff deflated and no upper airway obstruction exists. Commercially available

one-way valves vary in their airflow resistive properties^{464,472} and effectiveness for promoting understandable speech.⁴⁷³

Figure 40-6



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Passy-Muir one-way valves that allow articulated speech through the native airway for spontaneously breathing patients with a tracheotomy or patients undergoing mechanical ventilation through a tracheotomy tube. (Courtesy Passy-Muir Tracheotomy Valves, Passy-Muir Inc., Irvine, CA).

Weaning from Tracheotomy and Decannulation

Patients recovering from prolonged respiratory failure may either tolerate rapid decannulation or require stepwise tracheotomy weaning. The presence of underlying cardiorespiratory or neuromuscular disease increases the likelihood that a period of weaning will be required.

Evaluation for decannulation begins with an interdisciplinary evaluation after the patient demonstrates success with spontaneous breathing for 48 hours.⁴⁷⁴ Persistent difficulties with spontaneous ventilation through the tracheotomy tube may occur if the tube is malpositioned in the airway causing partial obstruction²⁵⁶ or has a narrow internal diameter. Unfortunately, guidelines do not exist to inform clinical practices and extensive practice variation occurs between centers in evaluating patients for decannulation.⁴⁷⁵ Moreover, standardized decannulation protocols akin to protocols for weaning patients from mechanical ventilation have not been adequately studied to determine their effectiveness.^{14,391,476,477} Extensive practice variation occurs, therefore, with weaning patients from tracheotomy.⁴⁷⁸

Essential criteria for considering patients for decannulation include level of consciousness, cough effectiveness, volume of airway secretions, oxygenation, effectiveness of swallowing, and ability to breathe spontaneously with the tracheotomy tube capped.⁴⁷⁹ The ability to manage secretions is assessed by deflating the tube cuff and observing the patient for clinical signs of aspiration or retained secretions.⁴⁸⁰ Capping the tracheotomy tube and observing the

patient during spontaneous respiration for 30 minutes allows evaluation for the presence of upper airway obstruction.^{481,482} Increased inspiratory effort, stridor, or dyspnea provide evidence that the patient might benefit from downsizing the tracheotomy tube⁴⁷⁷ or tracheoscopy to evaluate for vocal cord or subglottic abnormalities.⁴⁸² Some centers routinely examine the airway by fiber-optic endoscopy before decannulation.⁴⁸³ Difficulty with spontaneous breathing while the tracheotomy is capped in the absence of airway abnormalities may indicate ventilator muscle weakness and potential benefits from weaning toward cannulation with a capped tracheotomy tube supplemented by noninvasive face-mask ventilation.⁴⁸⁴

Patients with moderate airway secretions and severely compromised lung function may not tolerate spontaneous breathing with partial obstruction of their airway with a tracheotomy tube. Such patients may benefit from a tracheotomy plug, also termed *stoma stent* or *stoma maintenance device*, which frees the airway but allows airway suctioning and maintenance of the stoma track for reinsertion of a tracheotomy tube if needed (Fig. 40-7).⁴¹¹

Figure 40-7



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**B**

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A stoma stent can maintain patency of the stoma after tracheotomy tube decannulation for patients who require ongoing airway access for suctioning or replacement of a tracheotomy tube if failure of decannulation occurs. Some stents are more adaptable for either long-term use (**A**, Montgomery Long-Term Cannula, courtesy Boston Medical Products, Westborough, MA) or shorter-term use (**B**, TRACOE Stoma Button, courtesy TRACOE Medical, Frankfurt, Germany), but each has allied attachments to promote articulated speech or provide oxygenation with humidification.

After prolonged ventilation, the failure rate of decannulation ranges from 5% to 20%.⁴⁸⁵ Up to 40% of patients may require translaryngeal intubation with readmission to the ICU.⁴⁸⁵ Most patients fail within the first 4 to 24 hours with retained secretions representing the major factor.⁴⁸⁵

Patient Prognosis and Quality Improvement

Patients who require tracheotomy for respiratory failure have a worse prognosis as compared with other ventilator-dependent critically ill patients and experience substantial health care costs.⁴⁸⁶ They represent the most resource-consuming diagnostic categories for many acute care hospitals.^{487,488} Their hospital mortality rate ranges from 49% to 78% if they undergo tracheotomy in the ICU.^{6,489–491} Among those who survive hospitalization, 24%, 30%, and 42% are no longer alive at 100 days, 6 months, and 2 years after discharge, respectively,⁴⁸⁹ and the presence of a tracheotomy is an independent predictor of long-term functional debility⁴⁹² and mortality.⁴⁹³ If decannulation occurs before hospital discharge, 92% of patients are alive 1 year later,^{489,494–496} but lack of decannulation before transfer

from the ICU to a ward has an independent association with hospital mortality^{497,498} with highest mortality among patients with obesity or neurologic injuries. Among mixed medical and surgical patients who survive acute hospitalization after tracheotomy, 57% are successfully weaned and 30% decannulated before hospital discharge.⁴⁸⁹ At 1 year, 9% of patients who require prolonged mechanical ventilation have no functional dependency, 26% are alive with moderate dependency, and 65% are either alive with complete functional dependency or dead.⁴⁸⁶ Older patients with more comorbid conditions are more likely to require discharge to post-acute care facilities.⁴⁸⁶ Incremental costs per quality-adjusted life-year exceed \$100,000 in older age groups.⁴⁹⁹

Unfortunately, clinical outcomes are significantly worse than projected by patients' surrogates and physicians at the time they make the decision to proceed with tracheotomy.⁵⁰⁰ Efforts to develop predictive models to estimate long-term outcomes for critically ill patients being considered for tracheotomy have had variable results.^{388,498,501}

Unfortunately, specialist clinicians in the ICU demonstrate practice variation for tracheotomy care that often does not conform to best clinical practices^{310,502} and delays decannulation.⁵⁰³ Combined with increasing trends for earlier transfer of patients from the ICU to hospital wards and post-acute care hospitals, an increasing number of patients with a tracheotomy are managed outside of the ICU by providers with limited knowledge and skills in airway care.^{14,504,505} Some centers have approached this challenge to quality care by developing interdisciplinary teams that provide tracheotomy care both in the ICU and after transfer to a non-ICU setting.^{15,392,474,506–508} Studies are limited regarding the impact of tracheotomy teams on patient outcomes, but a recent meta-analysis reported that all existing studies found that tracheotomy teams decreased time to decannulation, length of stay, and adverse events.⁵⁰⁹

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